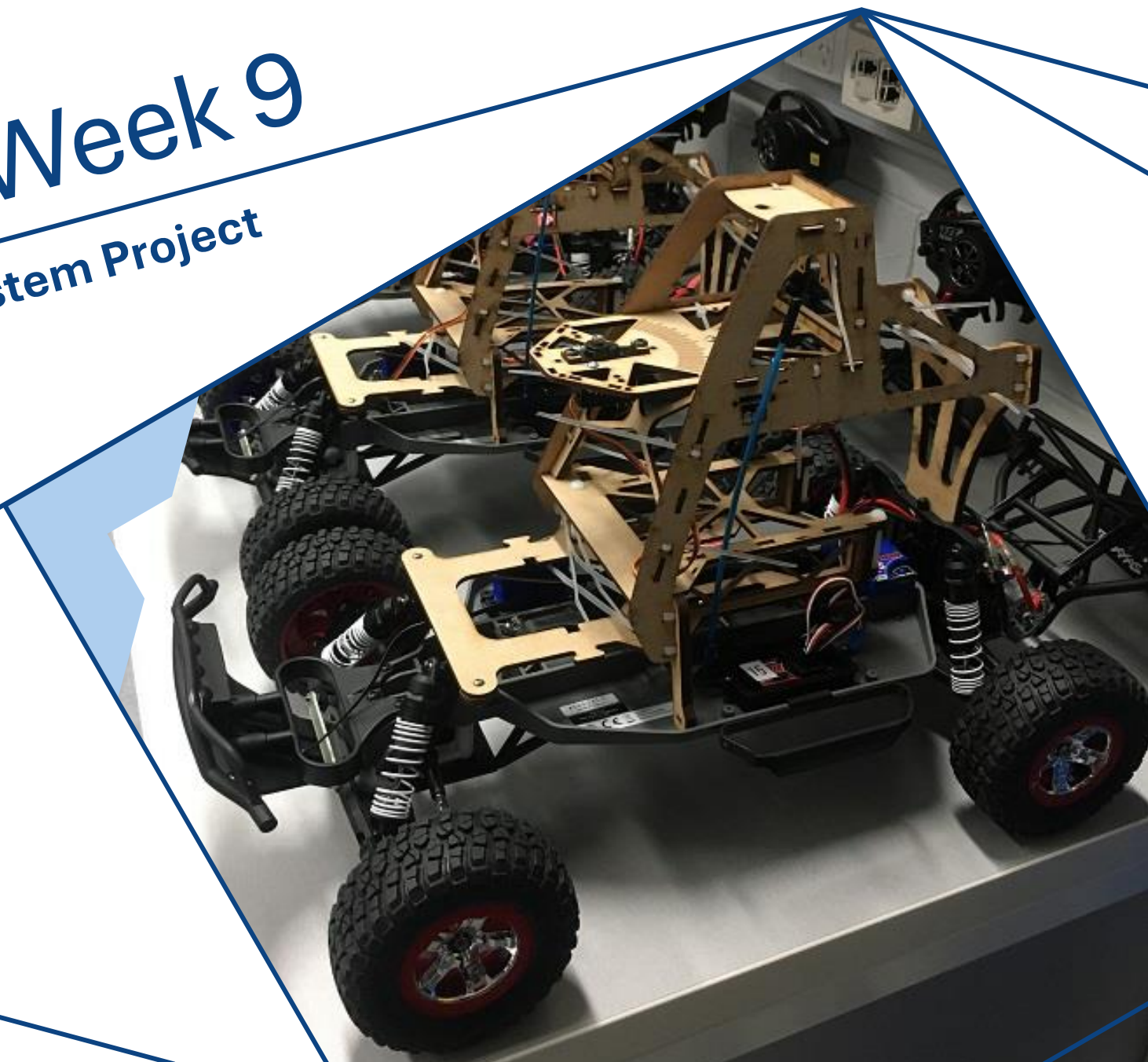


AI4R Workshop: Week 9

Team Formation Part 2 for the System Project

(Rev. 22 Sep 2026)

ELEN90095 AI-4-Robotics



The interdisciplinary flavour of ELEN90095

Exciting AI4R Tech

i.e., learning about architectures & algorithms that make robots do exciting things.

i.e., putting multiple building blocks together into a complete system (PID, RL, MPC, robustness, estimation, automata, ...).

Exciting AI4R Collaborations

i.e., embracing the opportunity to work with and exchange with others on a spectrum of AI4R topics.

i.e., developing and refining what it means for you personally to be an interdisciplinary team player on an AI4R project.

PHASE 1: Engage with AI4R Research (Week 1-3)

Read and present an AI4R paper.

Discuss the paper with peers.

PHASE 2: Individual Upskilling (Week 4-7)

Experience with PID, Deep RL, MPC

Share your experience with peers.

PHASE 3: System Project (Week 8-Exam Period)

Self-defined (and fun!) project

Working with peers as a team.

In the language of interdisciplinarity, this is...

Exploring the boundaries of:

- Knowledge, methods, problem formulations.
- Norms and language of technical details.
- Learning experiences.

In the language of interdisciplinarity, this is...

Developing the personal attributes of:

- Listening and entrepreneurship.
- System-level thinking.
- Reflective evaluation.

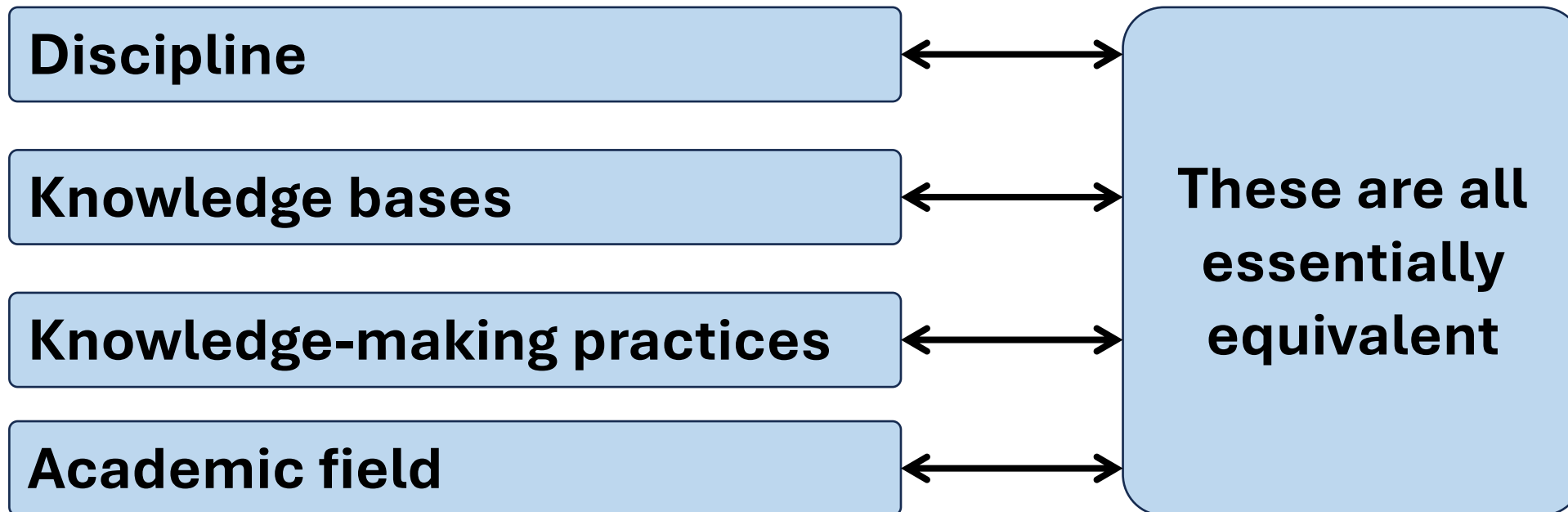
Interdisciplinary references

- The following references capture a range of “definitions” and perspectives on what interdisciplinary means for today’s world:
 - Kolmos, A., Holgaard, J. E., & Routhe, H. W. (2025). **Understanding and designing variation in interdisciplinary problem-based projects in engineering education.** *Education Sciences*, 15(2), 138. Link: <https://doi.org/10.3390/educsci15020138>
 - Kolmos, A., Holgaard, J. E., Routhe, H. W., Winther, M., & Bertel, L. (2023). **Interdisciplinary project types in engineering education.** *European Journal of Engineering Education*, 49(2), 257-282. Link: <https://doi.org/10.1080/03043797.2023.2267476>
 - Van den Beemt, A., MacLeod, M., Van der Veen, J., Van de Ven, A., Van Baalen, S., Klaassen, R., & Boon, M. (2020). **Interdisciplinary engineering education: A review of vision, teaching, and support.** *Journal of engineering education*, 109(3), 508-555. Link: <https://doi.org/10.1002/jee.20347>
 - Repko, F., R. Szostak, and M. P. Buchberger. (2019). **Introduction to Interdisciplinary Studies.** 3rd ed. Thousand Oaks: SAGE. Link: [to book on publisher’s website](#)
 - Schaffer, S. P., Chen, X., Zhu, X., & Oakes, W. C. (2012). **Self-efficacy for cross-disciplinary learning in project-based teams.** *Journal of Engineering Education*, 101(1), 82-94. Link: <https://doi.org/10.1002/j.2168-9830.2012.tb00042.x>
 - Lattuca, L. R., Knight, D. B., & Bergom, I. M. (2012). **Developing a measure of interdisciplinary competence for engineers.** In *2012 ASEE annual conference & exposition*, pp. 25-415. Link: <https://peer.asee.org/21173>

What is a discipline?

*“our working definition considers interdisciplinary interactions as attempts to address **real-world cases and problems** by integrating heterogeneous **knowledge bases and knowledge-making practices**, whether these are gathered under the institutional cover of a discipline or not”*

Van den Beemt et al (2020). **Interdisciplinary engineering education: A review of vision, teaching, and support**

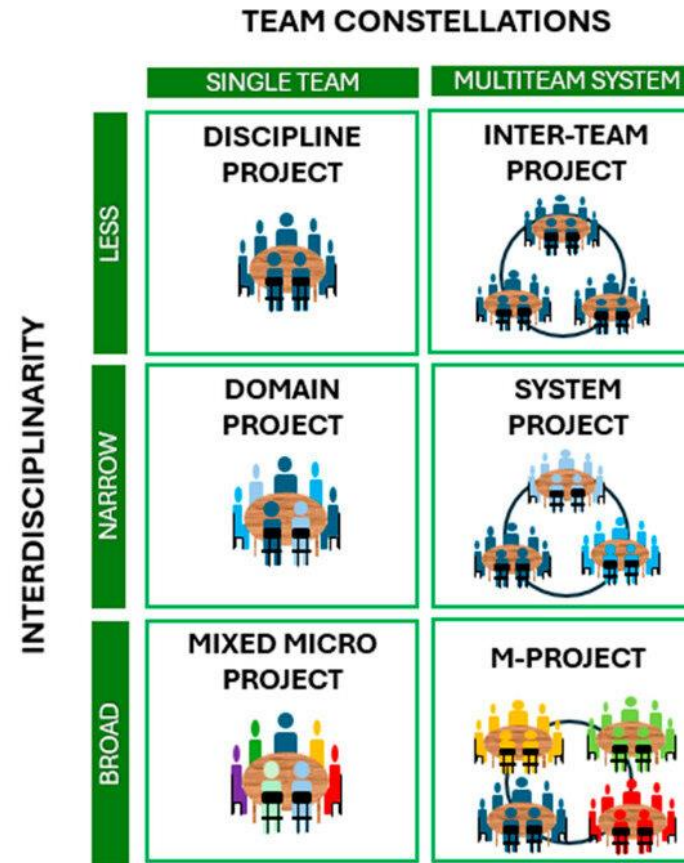


Extracts from references

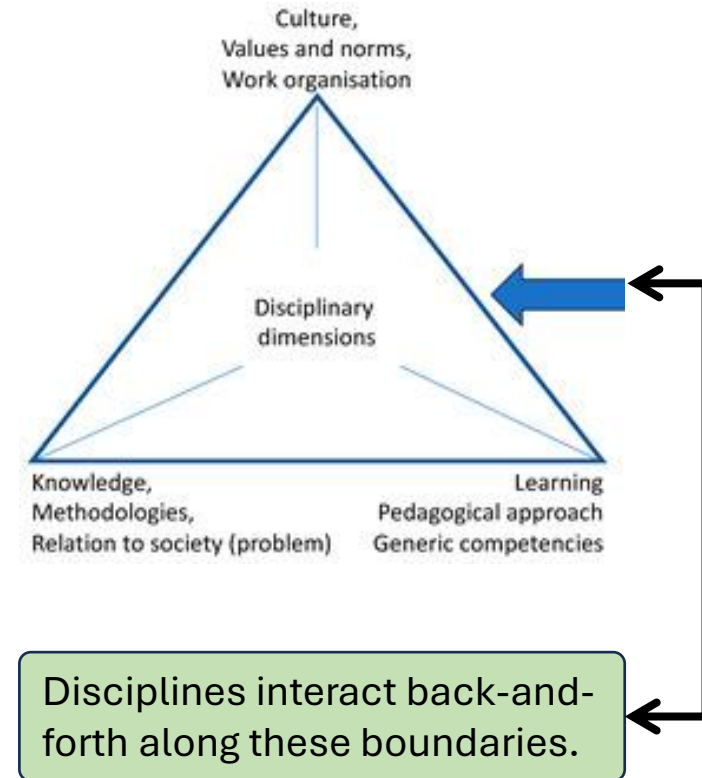
The paper describes that an interdisciplinary team requires the following traits **in every individual**:

1. An entrepreneurial mindset (i.e., **taking risks**).
2. A love of learning (i.e., being **excited to learn** something new).
3. Self-reflection (i.e., **awareness of one's own** strengths and weaknesses).
4. Intellectual courage (i.e., **acceptance of and respect** for other viewpoints).
5. Patience and empathy (i.e., **active listening**).

Repko et al. (2019). **Introduction to Interdisciplinary Studies**. 3rd ed.



Kolmos et al. (2023). **Interdisciplinary project types in engineering education**.



Kolmos, A. et al. (2025). **Understanding and designing variation in interdisciplinary problem-based projects in engineering education**.

Interdisciplinary vs. Multidisciplinary

In the language of **interdisciplinarity**, this is...

Exploring the boundaries of:

- Knowledge, methods, problem formulations.
- Norms and language of technical details.
- Learning experiences.

In the language of **interdisciplinarity**, this is...

Developing the personal attributes of:

- Listening and entrepreneurship.
- System-level thinking.
- Reflective evaluation.

In the language of **multidisciplinarity**, this is...

Divide and conquer

- Define the interfaces between the disciplines.
- Bring in the discipline experts for each part.

In the language of **multidisciplinarity**, this is...

Hierarchical team

- Teams are single discipline
- Team leaders figure out the “divide and conquer” interfaces.
- Team members do what they are told.

In reality, there are multiple spectrums for the different aspects of how teams of people from different disciplinary background work together.

Getting to know the cohort

ACTIVITY: complete the [Interdisciplinary Self-Efficacy Survey](#).

Team formation for the System Project

(Step 1) Finish forming a team

- Must meet the “Team Composition” requirements on the [System Project page](#).
 - Team size 4, 6, or 8 (or as close as possible to accommodate the class size).
 - At least 2, and preferably 3, different master's programs across the team members.
 - Maximum half the team members from the same master's program.
- Self-allocate into the same team on [Canvas People tab](#).
- Rename your team keeping the same first letter.

(Step 2) Complete the Getting Started Team Charter

- Create an MS Teams and create a Task board within your MS Teams team (see the [Teamwork Requirements page](#) for more details).
- Complete and submit the [Getting Started Team Charter as per the assignment on LMS](#).